

ROADEYE

RIDE EBYOND YOUR VISION

DESIGN MANUAL

Helmet Unit • Bike Module • Mobile Application

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1. Introduction

1.1 Overview

RoadEye is a smart motorcycle safety system designed to improve rider awareness and reduce road accidents caused by limited situational awareness, distraction, and delayed emergency response. The system combines a bike-mounted sensor module, a helmet-mounted Heads-Up Display (HUD) module, a companion mobile application, and a cloud backend to give riders real-time, hands-free, eyes-free assistance while riding.

1.2 Purpose of this Manual

This document serves as both a design manual and a user manual for RoadEye. It describes the overall system architecture and design decisions (for evaluators, maintainers, and future developers) and provides step-by-step instructions for installing, operating, and maintaining the device (for end users). Sections 2–4 focus on design and architecture; Sections 5–10 focus on installation, operation, and maintenance.

1.3 Problem Statement

Motorcyclists face a unique set of vulnerabilities on the road:

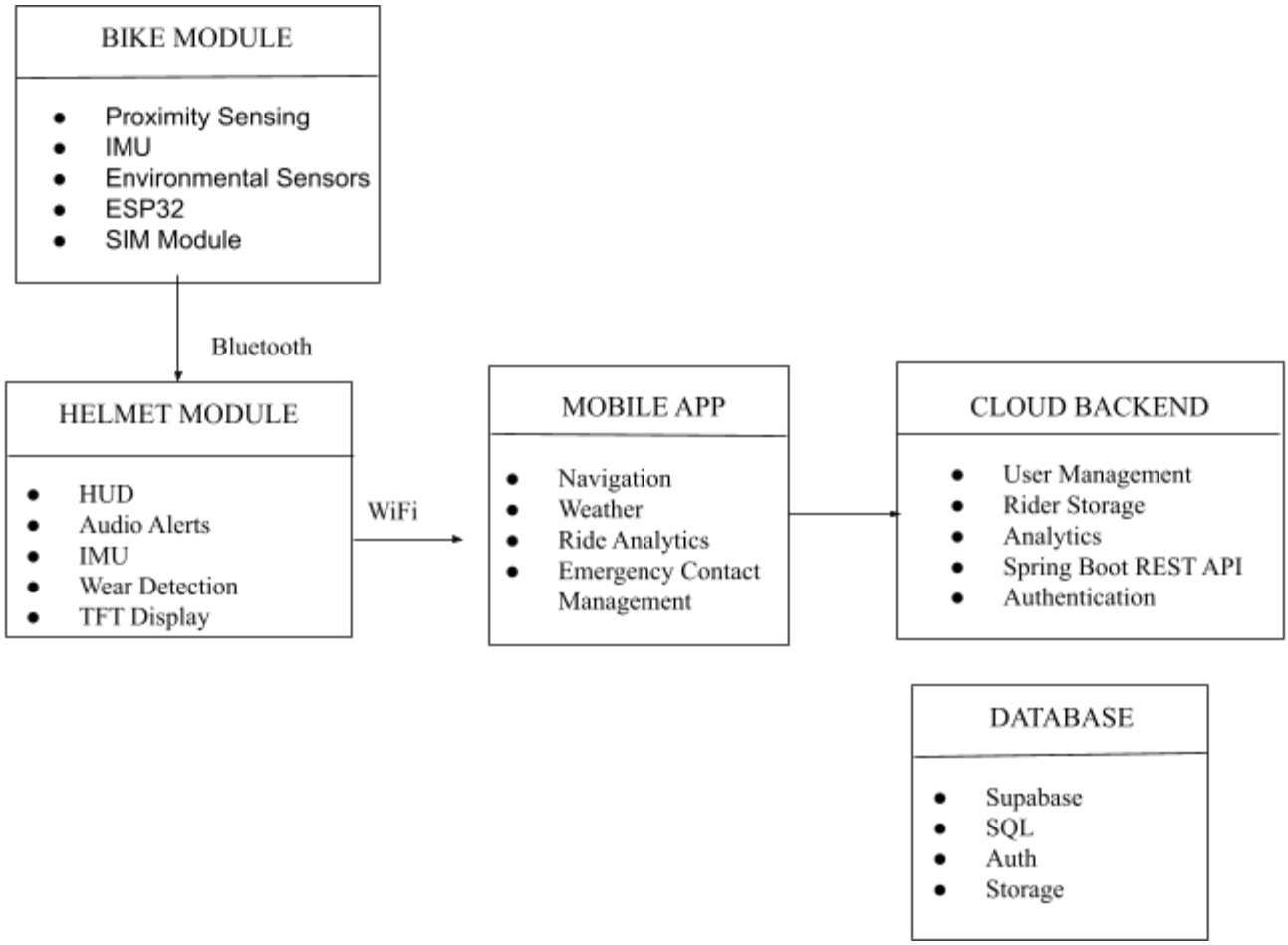
- Limited awareness of surroundings, particularly to the rear and sides.
- Existing navigation and communication tools are distracting and require the rider to look down at a phone or dashboard.
- Helmets themselves create blind spots that reduce peripheral vision.
- Emergency response after a crash is often delayed because no one is automatically notified.

1.4 Objectives

- Improve rider awareness without increasing mental load while riding.
- Detect dangerous situations (proximity, tilt, crash) in real time.
- Provide hands-free, eyes-free assistance through audio and a heads-up display.
- Collect and store ride statistics for the rider to review later.
- Automatically alert emergency contacts in the event of a crash.

2. System Architecture

RoadEye is composed of five major subsystems that communicate over Bluetooth and the internet: the Bike Module, the Helmet Module, the Mobile Application, the Cloud Backend, and the Database. The diagram below summarizes how data flows between them.



2.1 Bike Module

Mounted on the motorcycle itself, the Bike Module is responsible for sensing the vehicle's immediate environment and motion state. It is built around an ESP32 microcontroller and includes:

- Proximity sensing (ultrasonic sensors) for detecting nearby vehicles/obstacles.
- An Inertial Measurement Unit (IMU) for tilt and crash detection.
- Environmental sensors for temperature, humidity, and road/weather conditions.
- A SIM module for cellular connectivity, used for emergency alerts when a paired phone is unavailable.

2.2 Helmet Module

The Helmet Module is integrated into the rider's helmet and is responsible for presenting information to the rider and detecting rider-specific events. It includes:

- A Heads-Up Display (HUD) built around a small TFT display, optically refracted onto a lens in the rider's line of sight.
- Audio alerts via stereo speakers for turn-by-turn navigation, warnings, and emergency notifications.
- An IMU for head/helmet motion tracking.
- A capacitive wear-detection sensor that confirms the helmet is being worn.
- Haptic motors for silent, non-visual alerts (e.g., blind-spot warnings).

2.3 Mobile Application

The companion mobile app is the rider's control center before and during a ride. It handles user authentication, navigation, weather display, ride analytics, and emergency contact management, and it streams relevant information to the Helmet Module's HUD over a local wireless link.

2.4 Cloud Backend

A Spring Boot REST API, deployed on AWS, provides user management, ride-data storage, analytics, authentication, and crash-reporting services. It is designed to auto-scale with rider demand.

2.5 Database

RoadEye uses a PostgreSQL 15 relational database hosted on Supabase Cloud, accessed through a HikariCP connection pool for fast, concurrent access. This provides organized data structure, data consistency, automatic backups, and easy horizontal scaling.

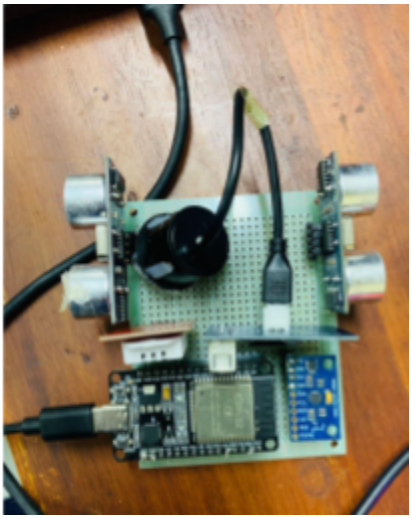
3. Hardware Design

3.1 Bike Module Hardware

The table below lists the sensors used in the Bike Module and the measurements they enable.

Sensor	Function	Measurement Enabled
HC-SR04 (×3)	Proximity detection	Distance to nearby vehicles
MPU6050	Tilt & crash detection	Tilt angle, sudden impact
BME680	Environmental sensing	Temperature, humidity, road/weather conditions



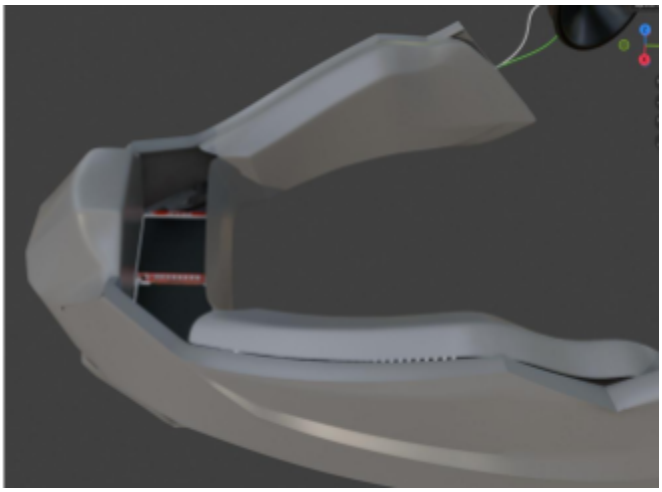


3.2 Helmet Module Hardware

The Helmet Module combines sensors that detect the rider's state with actuators that communicate information back to the rider.

Category	Component	Purpose
Sensor	IMU Sensor	Tracks helmet orientation and motion
Sensor	Capacitive Sensor	Detects whether the helmet is being worn
Actuator	TFT Display	Renders the HUD content (speed, navigation, alerts)
Actuator	Stereo Speakers	Delivers voice guidance and audio alerts
Actuator	Haptic Motors	Provides silent vibration-based alerts

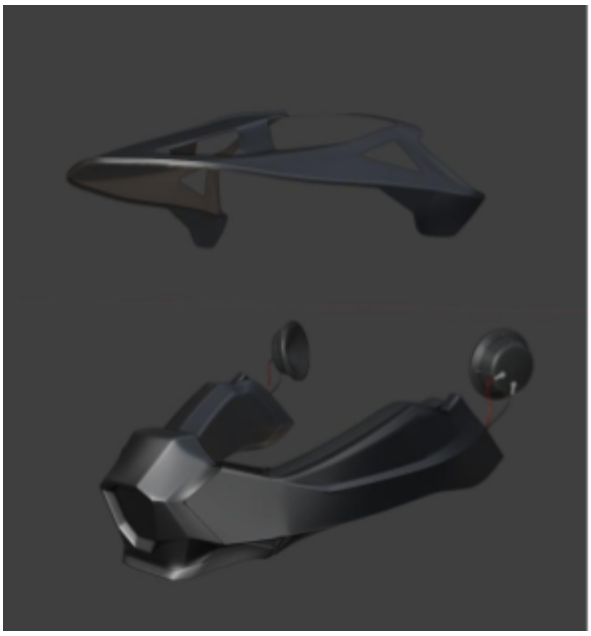




3.3 Helmet HUD Design

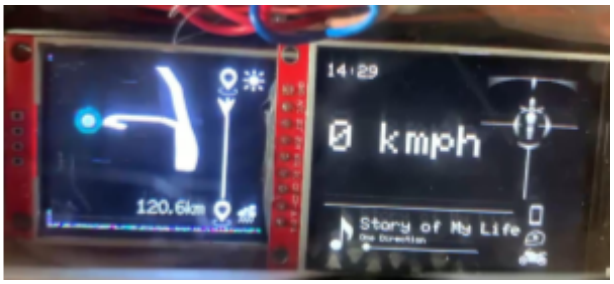
A key design challenge was presenting information to the rider without causing eye strain or distraction, since a display placed too close to the eye is hard to focus on comfortably.

Problem	Solution	Benefit
Near-display eye strain and distraction	TFT display refracted off a lens onto the rider's field of view	Safer information viewing; reduced eye accommodation effort









3.4 Safety-Oriented Helmet Mouthpiece

Adding electronic components to a helmet raises an important safety question: could the additional hardware increase injury risk during an accident? To address this, the team designed a custom fiberglass mouthpiece that is lightweight and impact-resistant, and that provides a secure, dedicated mounting point for the RoadEye components. This improves overall safety, durability, and integration with the base helmet shell.





3.5 Sensor Characteristics & Limitations

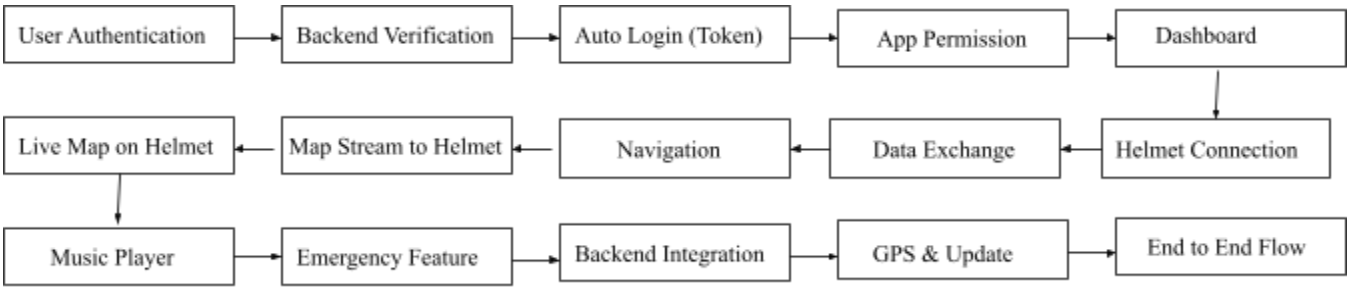
Every sensor used in RoadEye has known limitations that were identified during testing, along with the mitigation applied.

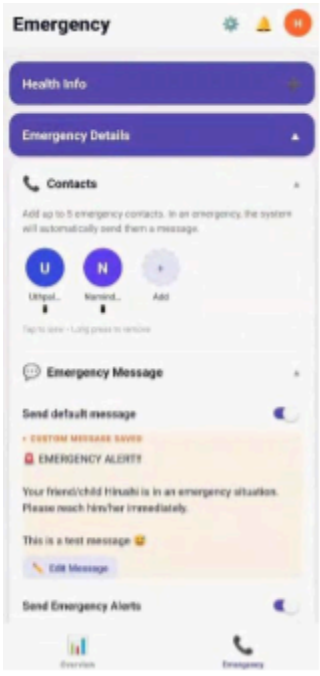
Measurement	Typical Rate	Known Limitation	Mitigation
Ultrasonic proximity	100 ms	Reflection/noise errors from irregular surfaces	Median window filter
IMU	50 ms	Gyroscope drift over time	Calibration and complementary filters
Crash detection	Real-time	Limited operating time due to battery constraints	Sleep/low-power modes when idle

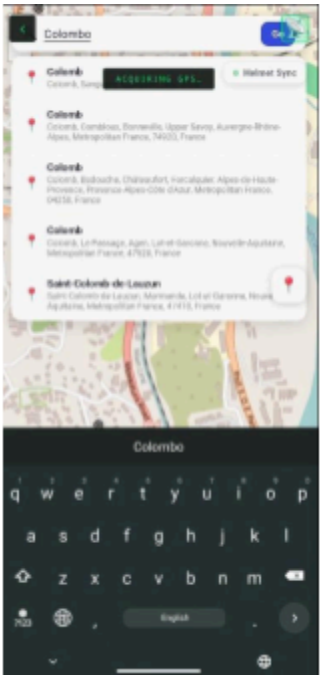
4. Software Design

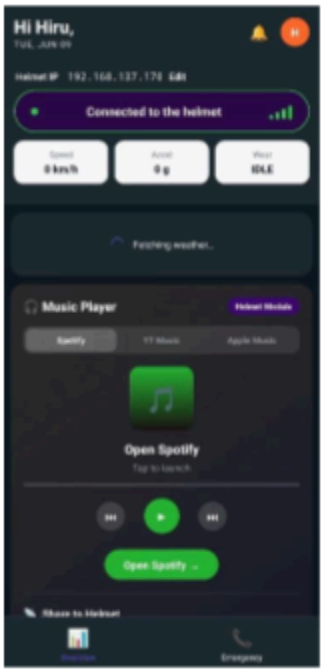
4.1 Mobile Application Flow

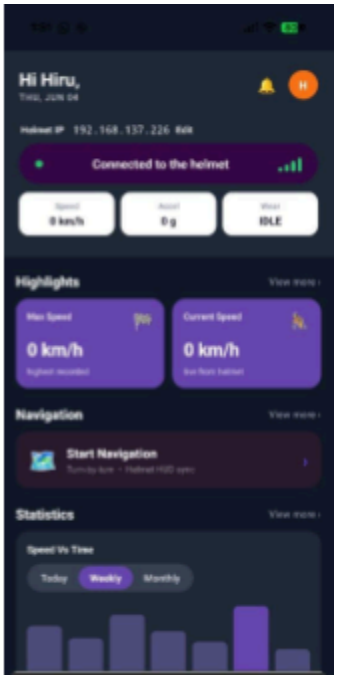
The mobile application follows a linear onboarding-to-ride flow: the user authenticates, the backend verifies credentials and issues a token for auto-login on future launches, the app requests the necessary device permissions, and the rider lands on the dashboard. From there, the app establishes a live map view, streams the map to the helmet, connects to the Helmet Module over Bluetooth/Wi-Fi, exchanges data continuously, and provides navigation, a music player, and emergency-contact features throughout the ride.





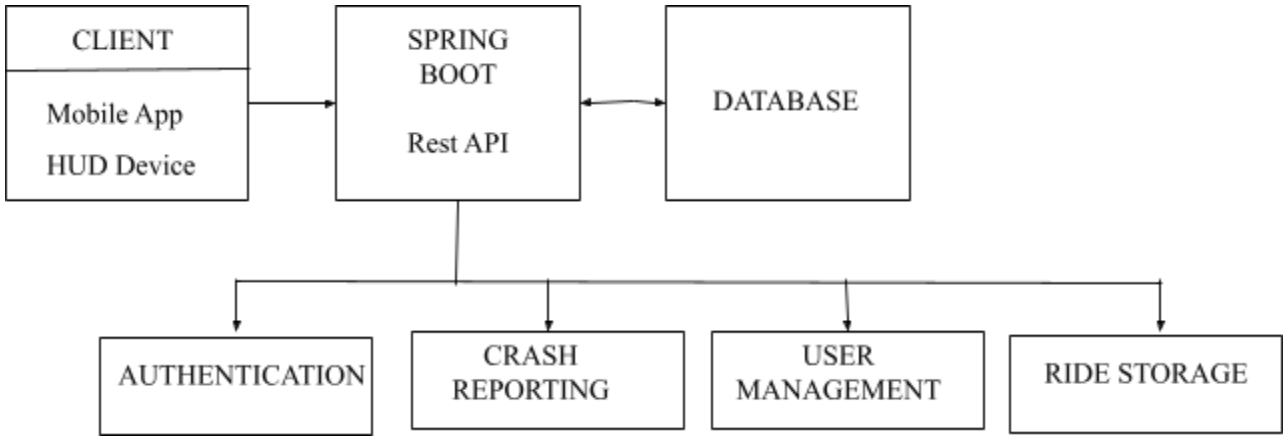






4.2 Backend Architecture

The backend is a Spring Boot REST API deployed on AWS. It exposes endpoints for authentication, user management, ride storage, and crash reporting, and communicates with the client (mobile app and HUD device) over HTTPS and with the PostgreSQL/Supabase database beneath it.



4.3 Database Design

Technology	Benefit
PostgreSQL 15 (relational database)	Secure data storage, complex queries, maintained relationships
Supabase Cloud database hosting	Automatic backups, high availability, secure cloud access
HikariCP connection pool	Faster API responses, handles many concurrent users, improved performance

Together these choices give the system an organized data structure, strong data consistency, faster queries, and straightforward scalability.

5. Installation & Setup Guide

5.1 What's in the Box

- 1 × Bike Module (ESP32 unit with proximity, IMU, and environmental sensors, and SIM module)
- 1 × Helmet Module (HUD unit, speakers, haptic motors, capacitive wear sensor, custom fiberglass mouthpiece)
- Mounting hardware and cabling
- USB charging cable(s)
- Quick-start card



5.2 Mounting the Bike Module

- Choose a dry, vibration-dampened location on the motorcycle frame (commonly under the seat or near the headlight assembly).
- Position the ultrasonic sensors with a clear, unobstructed line of sight to the front, left, and right.
- Secure the module using the supplied mounting bracket and cable ties.
- Connect the module to a stable power source (battery tap or auxiliary output), respecting polarity.

> ***⚠ Warning: Disconnect the motorcycle battery before making any electrical connections to avoid short circuits.***

5.3 Mounting the Helmet Module

- Fit the custom fiberglass mouthpiece to the interior of the helmet as per the shape of your helmet model.
- Attach the TFT display/HUD unit so that its reflected image falls within the rider's natural line of sight, just below eye level.
- Mount the stereo speakers near the ears and the haptic motors against the sides of the helmet liner.
- Confirm the capacitive wear-detection pad sits against the skin/liner contact area.

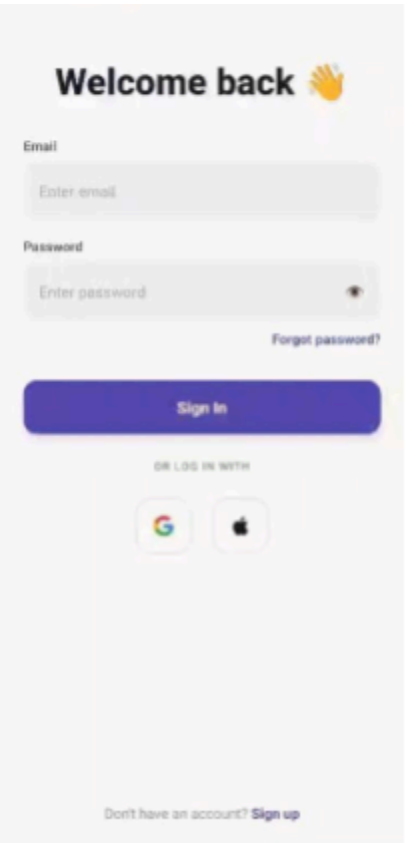
5.4 Pairing the Modules

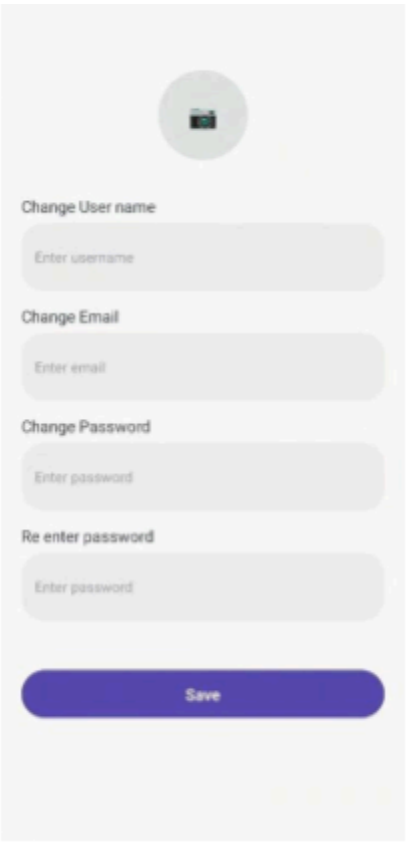
- Power on the Bike Module and the Helmet Module.
- The Bike Module automatically pairs with the Helmet Module over Bluetooth on first power-up.
- A confirmation tone / HUD message indicates a successful pairing.

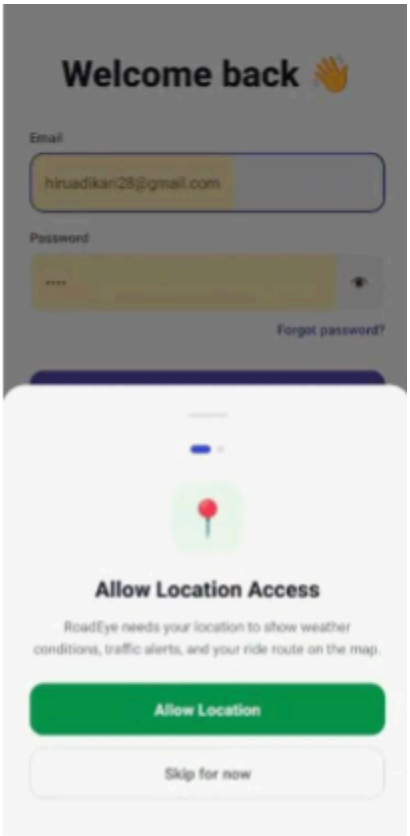
> *Note: If pairing fails, power-cycle both modules and ensure they are within 2 metres of each other during the initial pairing.*

5.5 Installing the Mobile Application

- Download the RoadEye app from your device's app store.
- Grant the requested permissions (location, Bluetooth, notifications) — these are required for navigation, helmet connectivity, and emergency alerts.
- Create an account or log in with existing credentials.
- Enable auto-login if you want the app to remember your session on future rides.







5.6 Connecting the App to the Helmet

- Open the RoadEye app and navigate to the Dashboard.
- Select 'Connect Helmet' and choose your Helmet Module from the list of nearby devices.
- Once connected, the live map and ride data will automatically stream to the HUD.

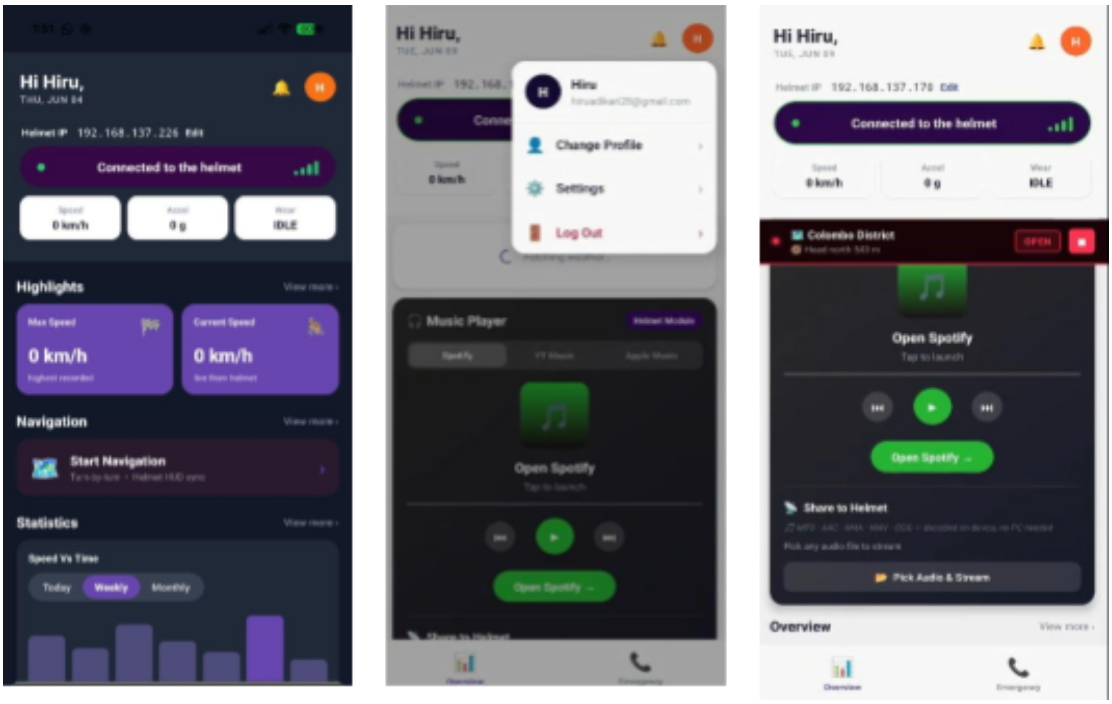
6. User Guide - Operating Instructions

6.1 Powering On

Press and hold the power button on the Bike Module until the status LED turns green. The Helmet Module powers on automatically once it detects the paired Bike Module, or can be switched on independently using its own button.

6.2 Dashboard Overview

The mobile app's dashboard is the main screen you will see after logging in. It shows your current ride status, quick access to navigation, weather, ride analytics, and emergency contact settings.



6.3 Using Navigation

- Enter your destination in the app's navigation field.
- Tap 'Start Ride' to begin streaming turn-by-turn directions to the helmet HUD.
- Voice guidance will also play through the helmet speakers.

6.4 Reading the HUD

While riding, the HUD displays your current speed, elapsed ride time, a compass/heading indicator, phone/helmet/motorcycle battery levels, currently playing track, and — when navigation is active — a turn indicator and remaining distance.

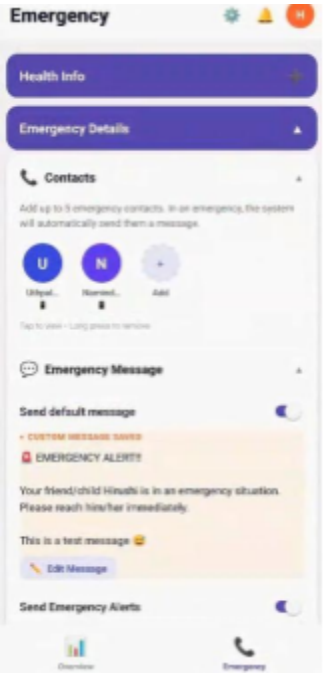
6.5 Music Player

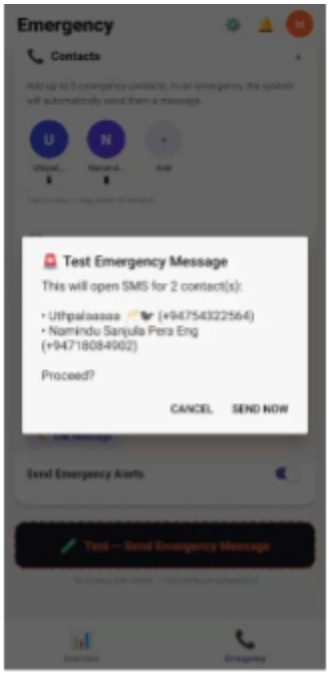
The currently playing track and playback progress are shown on the HUD and can be controlled via voice command or the handlebar-mounted control, if fitted, without needing to look at your phone.

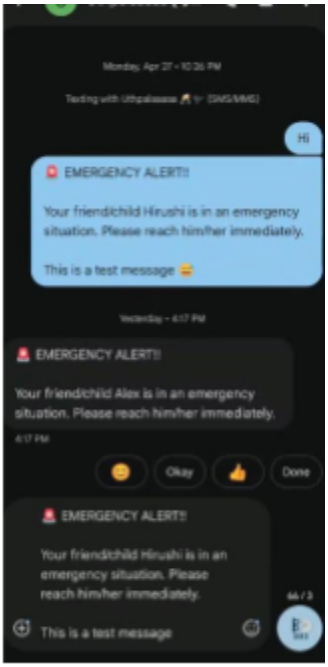
6.6 Emergency Features

The Bike Module's IMU continuously monitors tilt angle. If a sudden, sustained tilt beyond the configured threshold is detected — indicating a likely crash — the system automatically records the event and triggers the emergency workflow:

- The event (tilt angle, GPS coordinates, timestamp) is logged and sent to the backend.
- The mobile app is notified and starts a countdown, giving the rider a chance to cancel a false alarm.
- If not cancelled, an SMS/notification is sent to the rider's configured emergency contacts with the location of the incident.







6.7 Ride Analytics

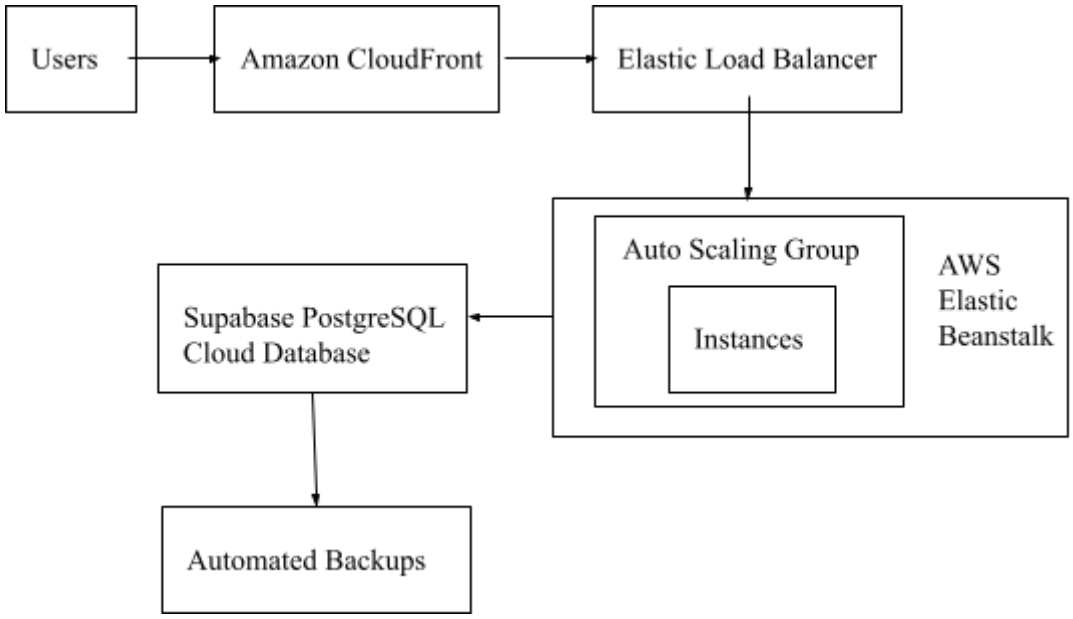
After each ride, RoadEye stores statistics such as distance travelled, average speed, and any recorded events. These can be reviewed at any time from the Ride Analytics section of the app.

7. Non-Functional Design Considerations

Aspect	Design Target / Approach
Dependability	99.5% uptime, automated backups, fault tolerance
Efficiency	HUD refresh under 60 ms; alert latency under 200 ms
Scalability	Auto-scaling cloud architecture supporting 50,000+ users
Security	End-to-end encrypted communication
User Experience	HUD-based interaction, voice guidance, haptic feedback

7.1 Scalability & Deployment

RoadEye's backend is deployed on AWS behind Amazon CloudFront and an Elastic Load Balancer, with an Auto Scaling Group (minimum 1, maximum 10 instances) running on AWS Elastic Beanstalk. Data is persisted in a Supabase-hosted PostgreSQL database with automated backups. This configuration is designed to support over 50,000 users.



7.2 Security Analysis

Because RoadEye handles sensitive data — passwords, user profiles, crash locations, emergency contacts, and push-notification tokens — the design accounts for a range of threats and mitigations.

Sensitive Data	Relevant Threats	Security Measures
Passwords, user profiles, crash locations, emergency contacts, push tokens	Data interception, unauthorized access, SQL injection, JWT theft, brute-force attacks	BCrypt password hashing, JWT authentication, TLS 1.3, Spring Security, input validation, role-based access control

8. Testing & Validation Summary

8.1 Software Testing

Test Type	Tools / Coverage	Result
Unit Testing	JUnit, Mockito — 85% coverage	Passed
Integration Testing	Authentication, Ride APIs, Crash APIs, Emergency notifications	All 28 endpoints tested successfully

8.2 Hardware Testing

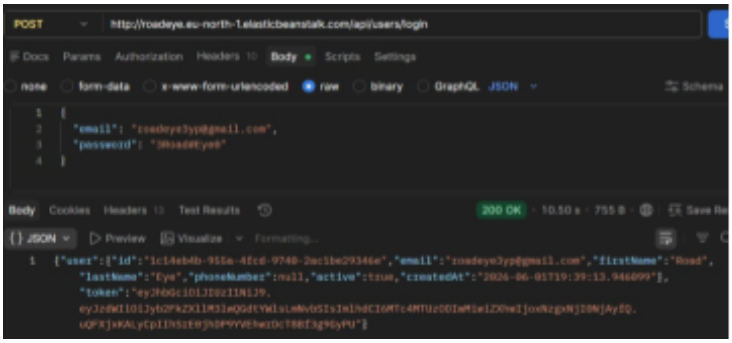
Each sensor was calibrated and checked against its expected accuracy:

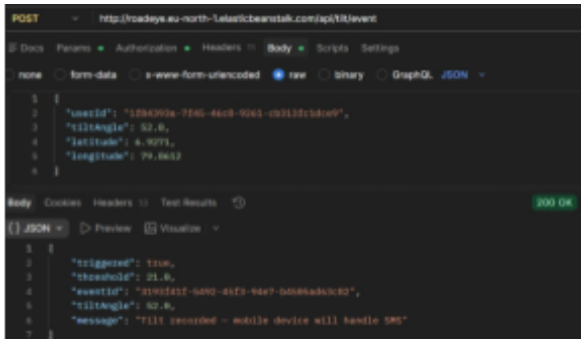
Sensor	Measured Accuracy
MPU6050 (IMU)	±3%
HC-SR04 (Proximity)	±2 cm
BME680 (Environmental)	±1°C

Additional functional tests were conducted for crash detection, BLE communication, HUD rendering, map streaming, and sensor calibration — all components were confirmed operational.

8.3 Testing Proofs

Example API tests, performed using Postman, verified both the login flow and the emergency (tilt) detection flow end-to-end against the deployed backend.





8.4 Functional Requirements Checklist

Requirement	Status
HUD	Complete
Navigation	Complete
Crash Detection	Complete
Emergency Alerts	Complete
User Authentication	Complete
Ride Tracking	Complete
Proximity Detection	Complete
Cloud Deployment	Complete
Mobile Application	Complete
Data Storage	Complete

9. Maintenance & Troubleshooting

9.1 Charging

- Charge the Bike Module and Helmet Module fully before first use.
- Battery levels for phone, helmet, and motorcycle modules are displayed on the HUD dashboard at all times.
- Avoid charging in direct sunlight or extreme heat.

9.2 Cleaning & Care

- Wipe the helmet module and mouthpiece with a soft, dry cloth; avoid solvents that could damage the fiberglass finish.
- Keep the ultrasonic sensors on the bike module free of dirt/mud for accurate proximity readings.

9.3 Firmware Updates

The app will notify the rider when a firmware update is available for the Bike or Helmet Module. Updates should be applied while the motorcycle is stationary and the modules are sufficiently charged.

9.4 Common Issues

Symptom	Possible Cause	Suggested Action
Helmet Module won't pair	Devices out of range or already paired to another unit	Power-cycle both modules and retry within 2 metres
Inaccurate proximity readings	Dirt/mud on ultrasonic sensors or reflective surfaces nearby	Clean sensors; readings are also smoothed with a median filter
HUD image not visible / faint	Lens misaligned or ambient light too bright	Re-adjust the HUD lens position; check visor tint
Crash detected falsely	Pothole or sudden but non-crash tilt event	Cancel the alert countdown in the app within the grace period
App shows 'disconnected'	Bluetooth/Wi-Fi link dropped or phone out of range of helmet	Ensure phone stays within range and Bluetooth is enabled

10. Safety Guidelines

> ***⚠ Warning: RoadEye is a rider-assistance tool. It does not replace safe riding practices, attentive riding, or compliance with local traffic laws.***

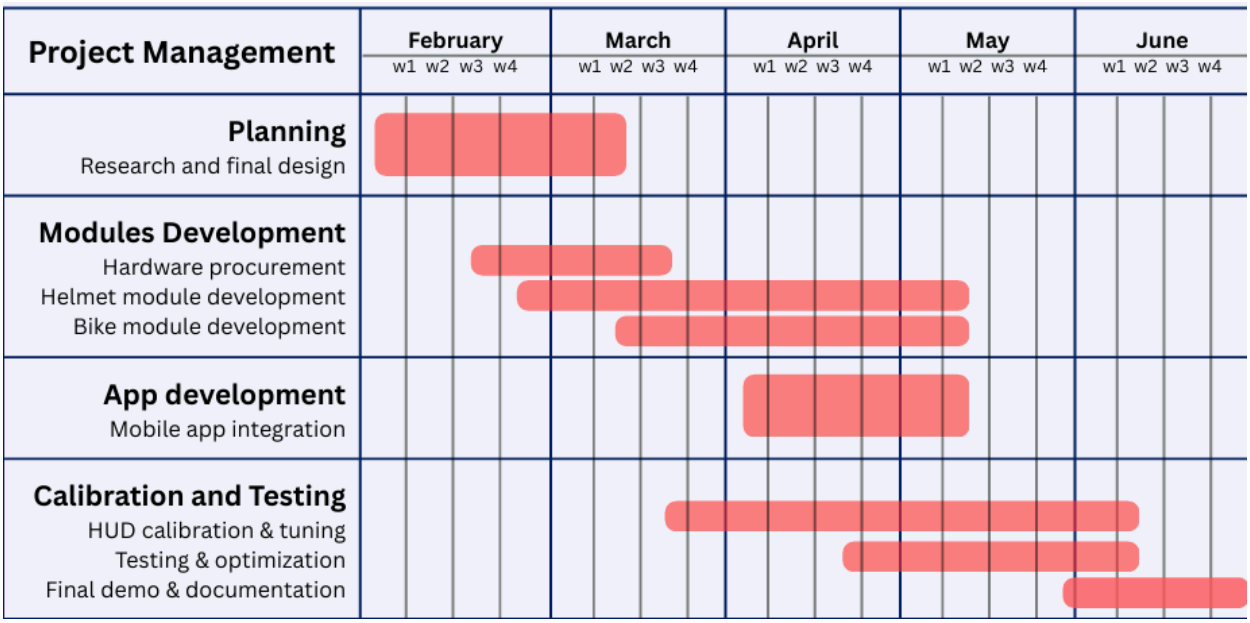
- Always ensure your helmet is DOT/ECE certified and properly fitted; RoadEye's mouthpiece and modules are designed to integrate with, not replace, standard helmet safety features.
- Do not attempt to adjust the HUD, navigation, or music player while the motorcycle is in motion — voice and haptic feedback are provided specifically so you do not need to.
- Do not rely solely on the emergency-alert feature; carry a phone and inform someone of your route when riding long distances or in remote areas.

- Have the electrical installation of the Bike Module inspected by a qualified technician if you are not confident performing it yourself.
- Replace or recharge batteries as directed; do not use damaged batteries or chargers.

11. Appendix: Development Roadmap

The table below summarizes the development phases followed in building RoadEye, from initial planning through to the final demonstration.

Phase	Activities	Approximate Timeline
Planning	Research and final design	February
Modules Development	Hardware procurement; helmet module development; bike module development	February – April
App Development	Mobile app integration	April
Calibration & Testing	HUD calibration & tuning; testing & optimization; final demo & documentation	April – June



— End of Document —